ANATOMY

Urinary Tract & Posterior Abdominal Wall

SGL 1 · Dr. Faris Muhammed
GUE BLOCK

Anatomy GUE Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

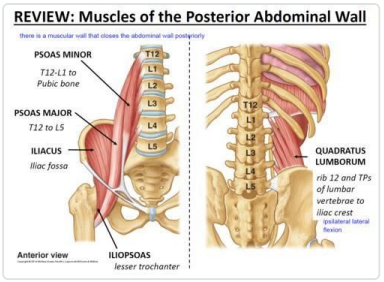
1. Name the three narrow points of the ureter where a stone is most likely to stick.
2. Which suprarenal vein drains into the IVC, and which drains into the left renal vein?
3. Which autonomic outflow contracts the detrusor and relaxes the internal urethral sphincter?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Posterior abdominal wall — boundaries and muscles

Definition	The **muscular and skeletal wall** forming the posterior boundary of the abdominal cavity
Boundaries	Above: **diaphragm** · below: **pelvic brim** · in front: abdominal cavity and contents · behind: **lumbar vertebrae** + posterior abdominal muscles

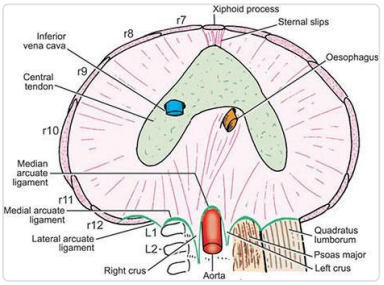
Muscle	Origin	Insertion
Psoas major	Bodies, IV discs and transverse processes of **T12–L5**	**Lesser trochanter** (via iliopsoas tendon)
Psoas minor	Bodies of **T12 and L1** + the disc between	**Pectineal line & iliopubic eminence** via iliac fascia
Iliacus	**Iliac fossa**, iliac crest, ala of sacrum	Lesser trochanter, **with psoas major** (= iliopsoas)
Quadratus lumborum	**Iliolumbar ligament** + posterior iliac crest	**Lower border of 12th rib** + TPs of L1–L4 · action: **ipsilateral lateral flexion**



Psoas minor, psoas major, iliacus (→ iliopsoas) and quadratus lumborum.

2 Diaphragm — what the posterior wall hangs from

Arcuate ligaments	**Median** → over the **aorta** (coeliac trunk arises just below) · **medial** → over **psoas major** · **lateral** → over **quadratus lumborum**
Crura	**Right crus longer** (down to L3) · left crus to L2 (crura slide)
Oesophageal opening	Slung by the right and left bundles of the **right crus** (~60%; a crossed variant in ~40%) · vagus nerves pass with the oesophagus
IVC opening	In the **central tendon**
Foramina	**Morgagni** = sternocostal triangle (superior epigastric a. & v.) · **Bochdalek** = lumbocostal triangle
Ligament of Treitz	From the **right crus** to the **duodenojejunal flexure** (superior and inferior components)



Median, medial and lateral arcuate ligaments; right crus longer than left.

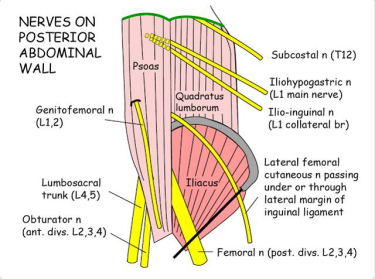
3 Nerves and vessels on the posterior wall

Nerve	Roots	Where it appears (slide figure)
Subcostal	**T12**	Lateral · uppermost, over quadratus lumborum
Iliohypogastric	**L1** (main nerve)	Lateral border of psoas, over QL
Ilio-inguinal	**L1** (collateral branch)	Lateral border of psoas, just below iliohypogastric
Genitofemoral	**L1, L2**	**Pierces psoas** → runs on its **anterior surface**
Lateral femoral cutaneous	—	Crosses iliacus → **under or through the lateral inguinal ligament**
Femoral	L2–4 **posterior** divisions	Lateral border, **between psoas and iliacus**
Obturator	L2–4 **anterior** divisions	**Medial** border of psoas

Nerve	Roots	Where it appears (slide figure)
Lumbosacral trunk	**L4, L5**	**Medial**, descends into the pelvis

> **Psoas is the landmark:** lateral border → iliohypogastric, ilio-inguinal, lateral femoral cutaneous, femoral · front → **genitofemoral** · medial border → **obturator** and lumbosacral trunk.

Arteries	**Abdominal aorta** · lumbar · common iliac · median sacral · renal
Veins	**Inferior vena cava** · lumbar · common iliac · renal

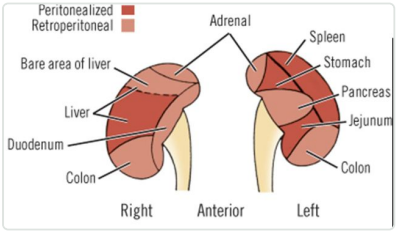


Lumbar plexus branches around psoas.

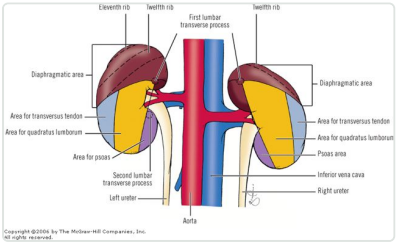
4 Kidney

Functions	Red cell production · **blood pressure** · blood pH (acid–base) · bone-mineral metabolism · **excretion** of waste and water
Size	**10 × 5.5 × 3 cm** (length × width × thickness)

	Right kidney	Left kidney
Front — peritoneum-covered	**Liver**	**Stomach, spleen, jejunum**
Front — retroperitoneal	Adrenal · bare area of liver · **duodenum** · colon	Adrenal · **pancreas** · colon
Behind (both)	Diaphragm · transversus tendon · **quadratus lumborum** · **psoas**	Same four areas
Ribs behind	**12th** only	**11th and 12th**
Transverse processes	L1	L1 (and L2 labelled)

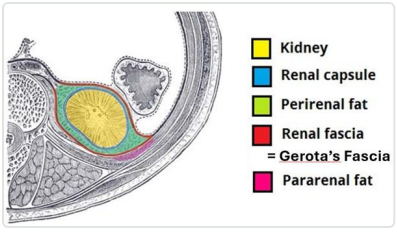


Anterior relations: dark = peritoneum-covered, light = retroperitoneal.



Posterior relations: left kidney 11th + 12th ribs, right 12th only.

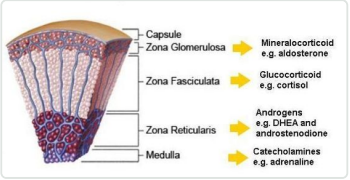
Inside → out	Renal papilla → **minor calyx** → **major calyx** → **renal pelvis** → ureter
Cortex / medulla	Medulla = **pyramids** · cortex dips between them as **renal columns (of Bertin)**
Arterial tree	Renal a. → **segmental** → **interlobar** → **arcuate** → **interlobular** · also inferior adrenal a., pelvic and ureteric branches
Segments	**Five**: apical, upper, middle, lower, posterior
Coverings (inner → outer)	Renal capsule → **perirenal fat** → **renal (Gerota's) fascia** → pararenal fat · Gerota's fascia also encloses the **adrenal**



Coverings: capsule, perirenal fat, Gerota's fascia, pararenal fat.

5 Suprarenal (adrenal) glands

Role	Part of the **hypothalamic–pituitary–suprarenal axis** · stress response and metabolism
Site	Yellowish, **retroperitoneal**, on the upper poles, **just above T12**
Wrapping	Inside the **renal fascia**, but separated from the kidney by **perirenal fat**
Structure	Outer **yellow cortex** · inner **dark brown medulla**



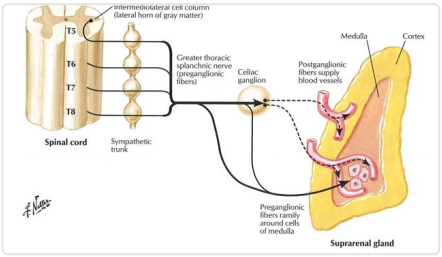
Cortex zones and medulla.

	Right gland	Left gland
Shape	**Pyramid** — caps the upper pole	**Crescent** — medial border, upper pole → hilum
In front	**Right lobe of liver + IVC**	**Pancreas, lesser sac, stomach**

	Right gland	Left gland
Behind	Diaphragm	Diaphragm
Vein (figure)	Right adrenal v. → **IVC**	Left adrenal v. → **left renal vein**

Layer	Makes	Example
Zona glomerulosa	Mineralocorticoid	**Aldosterone**
Zona fasciculata	Glucocorticoid	**Cortisol**
Zona reticularis	Androgens	DHEA, androstenedione
Medulla	Catecholamines	**Adrenaline**

Nerve supply	**Greater thoracic splanchnic nerve** (preganglionic; T5–T8 on the figure) ends **directly on medulla cells** — no synapse in the coeliac ganglion
Coeliac ganglion	Its postganglionic fibres supply the gland's **blood vessels**
Arteries (figure)	Branches from the **inferior phrenic arteries** above

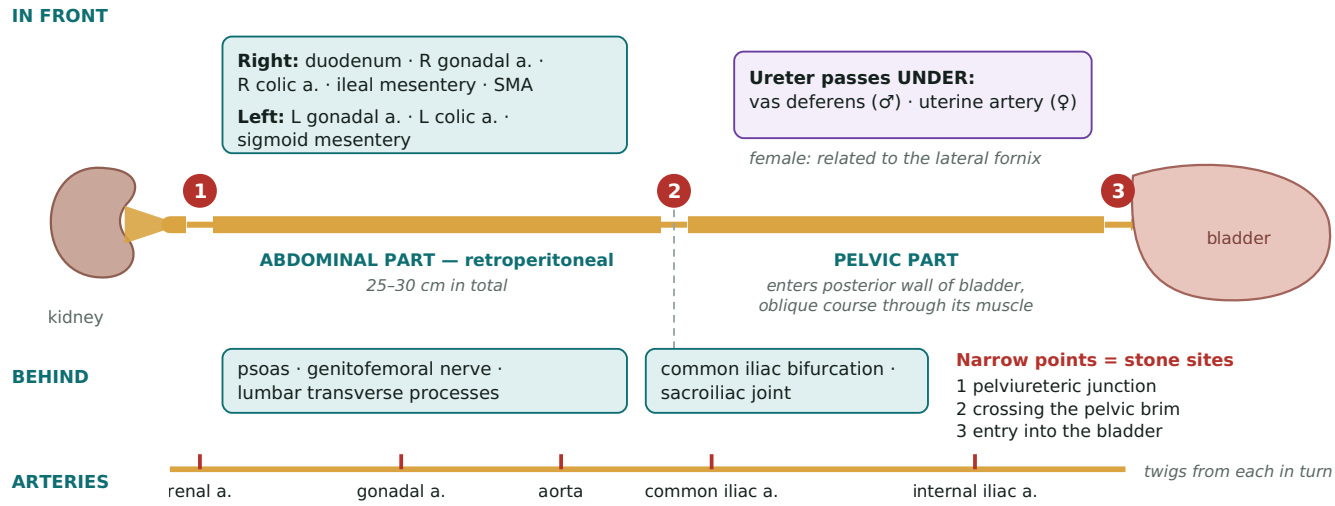


Preganglionic fibres ramify around medulla cells.

6 Ureter

Course	Continuous with the **renal pelvis** · behind the peritoneum · **in front of the lumbar transverse processes** · enters the **posterior wall of the bladder**
Length	**25–30 cm**
Into the bladder	Runs **obliquely through the muscle wall** before opening inside
Constrictions	**Three** — the same three sites where ureteric stones lodge (see diagram)

THE URETER'S JOURNEY — what lies in front, behind, and where it narrows

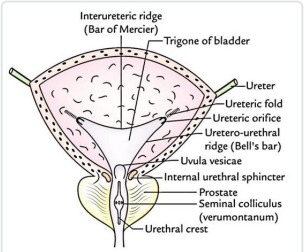


Arterial twigs come from each vessel the ureter passes on its way down.

Slide wording — "Passing under it: vas, uterine artery" — the figure shows the uterine artery crossing **above** the ureter, so this reads it as **the ureter passing under** both.

7 Urinary bladder

Nature / outside	**Collapsible muscular sac** — stores and expels urine · **apex**, superior surface, inferolateral surfaces · **superolateral angles** receive the ureters (vas deferens crosses here) · prostate below (male)
Trigone	Smooth triangle between the **two ureteric orifices** and the internal urethral orifice
Ridges	**Interureteric ridge = bar of Mercier** (joins the ureteric orifices) · uretero-urethral ridge = **Bell's bar** · **uvula vesicae** above the urethral opening



Trigone, bar of Mercier, Bell's bar, uvula vesicae.

Wall	Mucosa · **detrusor** muscle · adventitia · **internal urethral sphincter** at the neck
Male relations	In front: **retropubic space (of Retzius)** · behind: seminal vesicles, ampulla of ductus deferens, **rectovesical pouch**, rectum (rectoprostatic / Denonvilliers fascia between)

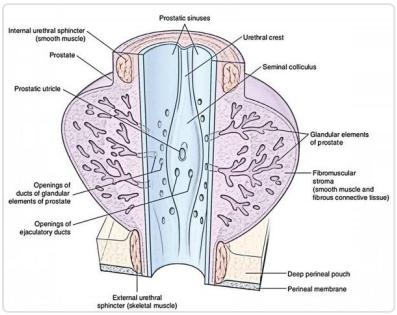
Nerve supply — read the + / − signs

Pathway	Roots → route	Detrusor	Internal sphincter	External sphincter
Sympathetic	**T11–L2** → hypogastric & vesical plexuses	**− relaxes**	**+ contracts**	—
Parasympathetic	**S2–S4** → pelvic splanchnic → vesical plexus	**+ contracts**	**− relaxes**	—
Somatic	**Pudendal nerve** (S2–S4)	—	—	**+ contracts**

Read across: sympathetic relaxes the wall and closes the outlet (**storage**); parasympathetic squeezes the wall and opens the outlet (**voiding**); the pudendal nerve gives voluntary control of the external sphincter.

8 Urethra

Length	Female **4 cm** · male **14 cm** (slide figures*)
Prostatic part	**Urethral crest** with the **seminal colliculus (verumontanum)** · on it: **prostatic utricle** + **ejaculatory duct openings** · prostatic ducts open into the **prostatic sinuses**
Sphincters	**Internal** = smooth muscle at the bladder neck · **external** = skeletal muscle (sphincter urethrae) around the next part
Intermediate (membranous)	Passes through the deep perineal pouch / perineal membrane · **bulbourethral (Cowper's) glands** beside it
Spongy (cavernous)	**Bulbous part** receives the bulbourethral ducts · **pendulous (penile) part**: urethral lacunae (Morgagni), glands of Littré · **navicular fossa** → external urethral orifice
Clinical (figures)	**BPH**: enlarged prostate narrows the prostatic urethra → obstructed outflow · **catheter** runs the whole urethra into the bladder



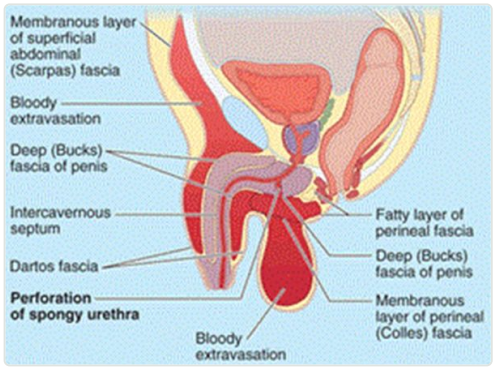
Prostatic urethra: crest, colliculus, utricle, duct openings, both sphincters.

*Many textbooks give a longer male urethra (around 18–20 cm). The lecture states 14 cm — check with the lecturer which figure is expected.

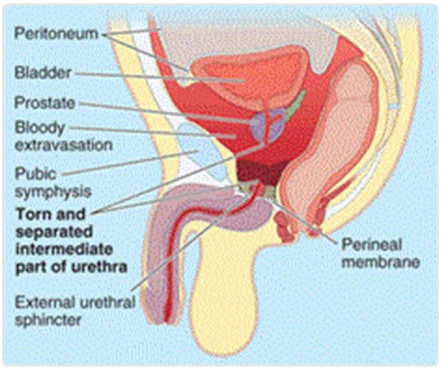
9 Urethral injury and extravasation

Fascial layers (figure)	**Scarpa's** fascia (abdomen) → **Colles'** (membranous perineal) fascia · **dartos** · **Buck's** (deep) fascia of the penis · deep perineal (Gallaudet) fascia · suspensory ligament of the penis

	Anterior urethral injury	Posterior urethral injury
Part torn	**Spongy** — bulbar or penile urethra	**Membranous** (intermediate) or prostatic urethra
Slide label	**Extra-abdominal**	"Intra-abdominal / **extraperitoneal**"
Where blood/urine goes	Deep to Colles'/Scarpa's fascia → **scrotum, penis, lower anterior abdominal wall**	**Above the perineal membrane**, around the prostate and bladder, below the peritoneum



Anterior: perforated spongy urethra.



Posterior: torn intermediate part above the perineal membrane.

P Must-remember pearls

- Ureter: **25–30 cm**, retroperitoneal, on psoas in front of the lumbar TPs · **three narrow points** (pelviureteric junction, pelvic brim, bladder entry) = stone sites.
- Kidney relations: right kidney meets the **liver** and **duodenum**; left meets **stomach, spleen, pancreas, jejunum** · ribs behind: right **12th**, left **11th + 12th**.
- Adrenal: right **pyramid** → vein to **IVC**; left **crescent** → vein to **left renal vein** · medulla gets **preganglionic** splanchnic fibres directly.
- Bladder: sympathetic **T11–L2** relaxes detrusor and closes the internal sphincter; parasympathetic **S2–S4** does the opposite; **pudendal** → external sphincter.
- Spongy urethra tear → scrotum, penis, abdominal wall · membranous tear → **above the perineal membrane**, extraperitoneal.

Q Test yourself

1. At cystoscopy the urologist follows a raised fold running between the two ureteric orifices. What is it called?

A. Uvula vesicae
B. Uretero-urethral ridge (Bell's bar)
C. Interureteric ridge (bar of Mercier)
D. Urethral crest
E. Seminal colliculus

2. On CT the coeliac trunk is seen arising from the aorta immediately below a fibrous arch of the diaphragm. Which arch?

A. Median arcuate ligament
B. Medial arcuate ligament
C. Lateral arcuate ligament
D. Central tendon
E. Left crus

3. A surgeon removing a right adrenal mass must first move which structures lying directly in front of the gland?

A. Pancreas and lesser sac
B. Stomach and spleen
C. Duodenum and right colic flexure
D. Diaphragm and 12th rib
E. Right lobe of the liver and the IVC

4. A perirenal fluid collection spreads up around the adrenal but stays inside the renal fascia. Which layer separates the adrenal from the kidney?

A. Pararenal fat
B. Perirenal fat
C. Renal capsule
D. Transversalis fascia
E. Parietal peritoneum

5. Which posterior-wall muscle runs from the iliolumbar ligament and iliac crest to the 12th rib and L1–L4 transverse processes, and bends the trunk to the same side?

A. Psoas major
B. Psoas minor
C. Iliacus
D. Quadratus lumborum
E. Transversus abdominis

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 C — The interureteric ridge — bar of Mercier — joins the two ureteric orifices; Bell's bar runs from each ureteric orifice to the urethra.

2 A — The median arcuate ligament arches over the aorta; the coeliac trunk arises just below it. Medial arches over psoas, lateral over QL.

3 E — Right gland: anterior — right lobe of liver and IVC; posterior — diaphragm. Pancreas, lesser sac and stomach lie in front of the left gland.

4 B — Both organs lie inside Gerota's (renal) fascia; the adrenal is separated from the kidney by perirenal fat.

5 D — Quadratus lumborum: iliolumbar ligament + posterior iliac crest → 12th rib + TPs L1–L4; ipsilateral lateral flexion.

SPACED REVIEW

☐ Day 1	☐ Day 3	☐ Day 7	☐ Day 21	☐ Pre-exam